

**Methodological details for variable construction for *Factors Influencing Recent Trends in Retail Electricity Prices in the United States***

y = starting year of timeframe

Y = ending year of timeframe

S = total in-state retail electricity sales in MWh

G = total in-state generation in MWh

| Variable                                                    | Description                                                                                                                                                                                                                                                                                                 | Methodological Details                                                                                                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| End-use-customer load delta [load delta]                    | Percentage change in state retail sales between starting year and ending year                                                                                                                                                                                                                               | $\frac{S_Y - S_y}{S_y}$                                                                                                                                    |
| Behind-the-meter solar delta [BTM delta]                    | Change in the amount of net metered behind-the-meter solar, as a percentage of state retail sales, between starting year and ending year                                                                                                                                                                    | $\frac{BTM_Y}{S_Y} - \frac{BTM_y}{S_y}$ BTM = BTM output in MWh                                                                                            |
| Utility-scale wind & solar share delta<br>wind+solar delta] | Change in the share of total in-state generation coming from utility-scale wind and solar between starting and ending year                                                                                                                                                                                  | $\frac{WS_Y}{G_Y} - \frac{WS_y}{G_y}$ WS = wind+solar output in MWh                                                                                        |
| RPS growth requirement [RPS growth]                         | Required growth in utility-scale renewable energy from starting year to ending year, beyond already-available RPS supply, as a percentage of state retail sales; multiplied by a regional estimate of the marginal incremental cost of renewable energy based on recent prices for renewable energy credits | $\left( \frac{RPS_Y}{S_Y} - \frac{RPS_y}{S_y} \right) \times REC$ RPS = RPS requirement in MWh<br>REC = marginal incremental cost (approximate REC prices) |

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| Natural-gas price exposure [gas exposure]                 | Average share of total in-state generation from natural gas from the starting year to ending year multiplied by the average price of gas over that same period at hubs from which gas may be purchased                                                                    | $\frac{\sum_{j=y}^Y \frac{gas_j}{G_j} \times gas\ price_j}{J}$ <p>gas = gas generation in MWh<br/> gas price = gas price (\$/MWh)<br/> J = # of years from y to Y</p> |
| Customer electric reliability change [reliability delta]  | Change in System Average Interruption Frequency Index (SAIFI) from starting year (and prior, assuming possible lagged effects) to ending year (and prior), focused solely on major events; SAIFI measures the average number of times customers experience a power outage | $SAIFI_Y + SAIFI_{Y-1} - SAIFI_y - SAIFI_{y-1}$ <p>SAIFI = System Average Interruption Frequency Index</p>                                                            |
| California dummy [California]                             | Binary variable for California, to reflect unique impact of wildfire risk on retail prices and any other unique aspects of the state                                                                                                                                      | 1 if state is California, 0 otherwise                                                                                                                                 |
| Prior price trend [prior trend]                           | Change in average retail price in the state in the 5-year period prior to the analysis period                                                                                                                                                                             | $p_y - p_{y-5}$ <p>p = retail electricity price</p>                                                                                                                   |
| Advanced Metering Infrastructure share growth [AMI delta] | Change in the share of advanced meters as a percentage of total meters from starting year (and prior, assuming possible lagged costs) to ending year (and prior)                                                                                                          | $\frac{AMI_Y + AMI_{Y-1}}{M_Y + M_{Y-1}} - \frac{AMI_y + AMI_{y-1}}{M_y + M_{y-1}}$ <p>AMI = # of advanced meters<br/> M = # of all meters</p>                        |
| Coal retirement amount [coal retirements]                 | Total capacity of coal plants retired in state between                                                                                                                                                                                                                    | $\sum_{j=1}^J \frac{CR_j}{G_j}$                                                                                                                                       |

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|                                                | starting year and ending year, divided by total in-state generation                                                                                                                                                                                                       | CR = MW of coal retirements                                                                                               |
| Demand response program size [DR size]         | Average demand response program size in state in terms of potential demand savings (MW) divided by total retail sales from starting year to ending year                                                                                                                   | $\frac{\sum_{j=y}^Y \frac{DR MW_j}{S_j}}{J}$ <p>DR MW = potential demand savings in MW<br/>J = # of years from y to Y</p> |
| Energy efficiency savings [EE delta]           | Change in estimated MWh savings from ratepayer-funded energy efficiency programs normalized by in-state retail electricity sales                                                                                                                                          | $\frac{EE_Y}{S_Y} - \frac{EE_y}{S_y}$ <p>EE = EE savings in MWh</p>                                                       |
| Electric vehicle share growth [EV delta]       | Change in the in-state share of all light duty vehicles that are EVs between starting and ending year                                                                                                                                                                     | $\frac{EV_Y}{V_Y} - \frac{EV_y}{V_y}$ <p>EV = # of electric vehicles<br/>V = # of all vehicles</p>                        |
| Competitive wholesale market [ISO]             | Binary variable for whether the state's utilities are part of a competitive wholesale electricity market, i.e., an ISO (in cases where a state is partly served by an ISO, the binary variable is coded as 'yes' if the majority of the state's load is served by an ISO) | 1 if state is in ISO, 0 otherwise                                                                                         |
| Publicly owned utility share [POU%]            | Average fraction of state retail sales served by a publicly owned utility from starting year to ending year                                                                                                                                                               | $\frac{\sum_{j=y}^Y \frac{POU_j}{S_j}}{J}$ <p>POU = POU sales in MWh<br/>J = # of years from y to Y</p>                   |
| Retail competition amount [retail competition] | Average fraction of state retail sales served by a competitive retailer from starting year to ending year                                                                                                                                                                 | $\frac{\sum_{j=y}^Y \frac{CRS_j}{S_j}}{J}$ <p>CRS = competitive retail supplier sales in MWh</p>                          |

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|                                                            |                                                                                                                                     | J = # of years from y to Y                                                                            |
| Natural hazards damage risk [risk: natural hazards]        | Expected annual loss from all natural hazards at the state level due to estimated damage to buildings, populations, and agriculture | $\frac{H}{S_Y}$ H = expected annual loss, all hazards                                                 |
| Wildfire damage risk [risk: wildfire]                      | Expected annual loss from wildfires at the state level due to estimated damage to buildings, populations, and agriculture           | $\frac{WF}{S_Y}$ H = expected annual loss, wildfires                                                  |
| Utility ownership of generation assets [utility owned gen] | Average fraction of all in-state generation capacity that is owned by an electric utility from starting year to ending year         | $\frac{\sum_{j=y}^Y \frac{UO_j}{G_j}}{J}$ UO = utility-owned generation<br>J = # of years from y to Y |